OFFICIAL PROJECT DOCUMENTATION

Integrated Project: Access to Drinking Water

Project Type

Data Analytics / Exploratory Data Analysis

Primary Tool

Google Sheets

Project Stage

Module 6 — Projects / Integrated Project

Dataset

Access to Drinking Water — 2020

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1. PROJECT OVERVIEW

Project Purpose

This project investigates access to drinking water across countries and explores how water-service levels differ according to:

- Population size
- Urban and rural population structure
- Income group

The project also examines the quality and distribution of the data before drawing conclusions.

The overall analytical journey moved from:

Data validation

→ **Population validation**

→ **Population structure**

→ **Water-access distribution**

→ **Population-size investigation**

→ **Income-group investigation**

→ **Cross-sectional comparison and interpretation**

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2. OVERALL BUSINESS / ANALYTICAL PROBLEM

The central problem explored in the project is:

> **How is access to drinking water distributed across countries, and what factors appear to be associated with differences in access?**

The analysis particularly investigates whether differences in water access appear to be associated with:

- Country population size
- Urbanization
- Rural population share
- Income group

The analysis is descriptive and exploratory. It identifies patterns and relationships but does not, by itself, establish causation.

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3. DATASET UNDERSTANDING

The dataset contains approximately 16 features per country/observation.

Important variables included:

Population

`pop_n`

National population, represented in thousands.

`pop_u`

Urban population share (%).

`pop_r`

Rural population share (%), derived during the project.

Income

`income_group`

Income category.

National Water Access

`wat_bas_n` — Basic water

`wat_lim_n` — Limited water

`wat_unimp_n` — Unimproved water

`wat_sur_n` — Surface water

Urban Water Access

`wat_bas_u`

`wat_lim_u`

`wat_unimp_u`

`wat_sur_u`

Rural Water Access

`wat_bas_r`

`wat_lim_r`

`wat_unimp_r`

`wat_sur_r`

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4. DATA VALIDATION AND PREPARATION

Before analysing water access, the integrity of the dataset was checked.

This was important because statistical analysis and visualization are only as reliable as the data being analysed.

4.1 Validating the Number of Values Per Row

A new feature called:

`value_cnt`

was created using:

```
=COUNTA(range)
```

Why COUNTA?

The dataset contained both:

- Numerical values
- Text values

`COUNT()` counts numerical values only.

`COUNTA()` counts both text and numerical values.

Since the expected dataset contained **16 columns**, correctly imported rows should contain:

> **16 non-empty values**

Rows containing only **15 values** were isolated.

4.2 Investigating Incorrectly Imported Rows

The affected rows contained instances where two intended values had been imported into the same cell and separated by a semicolon.

The correction process was:

1. Filter rows where `value_cnt = 15`.
2. Identify the cell containing the incorrectly combined values.
3. Move the misplaced value into the correct adjacent cell.
4. Use:

Data → Split text to columns

1. Use the semicolon as the delimiter.
2. Remove the filter.
3. Recheck `value_cnt`.

Result

The affected rows were corrected and returned to:

> **16 values per row**

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5. GLOBAL POPULATION VALIDATION

A new sheet called:

> **Global 2020 Report**

was created to summarize key calculations and findings.

The first analytical question was:

> **Does the dataset adequately represent the world's population in 2020?**

5.1 Dataset Population

The national population column `pop_n` was summed.

Result:

> **7,786,713,708 people**

5.2 World Population Reference

The project provided the estimated 2020 world population:

> **7,821,000,000 people**

5.3 Population Difference

The difference was calculated as:

Dataset population – World population

Result:

> **−34,286,292 people**

Using the absolute value:

```
=ABS(value)
```

the difference becomes:

> **34,286,292 people**

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6. PERCENTAGE DIFFERENCE

Percentage difference was used to provide context for the absolute population difference.

The formula was:

∣V2−V1∣V1+V22×100\frac{|V_2-V_1|}{\frac{V_1+V_2}{2}}\times100

Result

> **Approximately 0.44%**

Interpretation

Although the absolute difference represents more than 34 million people, that difference is only about **0.44%** of the average of the two population estimates.

This demonstrates an important analytical principle:

> **A large absolute number does not necessarily represent a large relative difference.**

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7. DATASET POPULATION COVERAGE

The percentage of the estimated world population represented by the dataset was calculated as:

```
=(Dataset Population / World Population)*100
```

Result:

> **99.56%**

Interpretation

The dataset represents approximately **99.56% of the estimated world population for 2020**.

This provided strong evidence that the dataset had extensive population coverage and was suitable for the subsequent exploratory analysis.

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8. URBAN AND RURAL POPULATION STRUCTURE

The dataset provided the urban population as a percentage share.

A rural population share was derived.

Formula

```
=100-pop_u
```

Therefore:

> **Urban share + Rural share = 100%**

This allowed the project to examine the population structure of each country.

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9. CONVERTING POPULATION TO MILLIONS

The project used the national population feature `pop_n`, which is stored in thousands.

Therefore:

> **1 unit of `pop_n` = 1 thousand people**

For example:
`1.35`

means:

> **1.35 thousand = 1,350 people**

To convert `pop_n` into millions:

```
=pop_n/1000
```

For example:

`53,771.3 ÷ 1,000 = 53.7713 million`

The project then created:

> `pop_n (m)`

using:

```
=ROUNDUP(C2/1000,0)
```

where appropriate for visualization grouping.

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10. IMPORTANT DATA REPRESENTATION LESSON

The population transformation created an important analytical distinction.

The transformation from:

> 53,771,300

to:

> 54 million

does **not** change the underlying population.

It changes the way the value is represented.

This was done to improve visualization readability rather than to replace the original population value.

Professional practice

The original population should be retained for precise reporting.

The rounded population feature can be used for:

- Grouping
- Visualization

- High-level comparisons

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11. INVESTIGATION 1 — ACCESS BY AREA

Objective

Investigate access to water across:

- National
- Urban
- Rural

populations.

The analysis focused on four service levels:

- Basic
- Limited
- Unimproved
- Surface water

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12. NATIONAL WATER-ACCESS DATA QUALITY CHECK

The maximum of each national water-access feature was calculated.

Feature	Maximum
Basic water	**100.0000037%**
Limited water	**37.42696287%**
Unimproved water	**33.53911377%**
Surface water	**30.36979308%**

Data Quality Finding

Basic water exceeded 100% by a very small amount.

A percentage cannot logically exceed 100%.

The observed value:

> **100.0000037%**

was treated as an erroneous precision/rounding issue.

---

13. CORRECTING BASIC WATER VALUES

A new feature was created:

> `wat_bas_n (rounded)`

Formula:

```
=ROUND(E2)
```

Why ROUND instead of ROUNDUP?

`ROUND()` applies normal mathematical rounding.

For example:

- 98.4 → 98
- 98.5 → 99
- 98.9 → 99

`ROUNDUP()` always moves upward.

For example:

- 98.1 → 99
- 100.0000037 → 101

Therefore, `ROUNDUP()` would introduce a larger error in this situation.

The use of `ROUND()` preserved the intended value of approximately 100%.

---

14. HANDLING `#VALUE!` ERRORS

Some observations produced:

> `#VALUE!`

when the rounding formula was applied.

This occurred because some original entries were non-numeric/missing values rather than numerical percentages.

These entries were handled as missing/non-numeric observations so they would not interfere with subsequent statistical calculations.

Analyst Lesson

A missing value and a genuine zero are not the same thing.

A value of:

> **0%**

can be a valid observation.

A missing/NAN entry means:

> The value is unavailable or not usable.

---

15. MINIMUM WATER-ACCESS VALUES

Feature	Minimum
Basic water	**37%**
Limited water	**0%**
Unimproved water	**0%**
Surface water	**0%**

Interpretation

The lowest observed basic-water access was:

> **37%**

while the other three water-service categories each reached:

> **0%**

This indicates that some observations recorded no population in the limited, unimproved, or surface-water categories.

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16. MEASURES OF CENTRAL TENDENCY

The following were calculated:

- Mean
- Median
- Mode

Results

Feature	Mean	Median	Mode
Basic water	**89.87%**	**97%**	**100%**
Limited water	**3.87%**	**0.47%**	**0%**
Unimproved water	**4.42%**	**0.85%**	**0%**
Surface water	**1.92%**	**0%**	**0%**

17. INTERPRETATION OF CENTRAL TENDENCY

Basic Water

Mean = 89.87%

Median = 97%

Mode = 100%

The high median and mode indicate that high basic-water access is common in the dataset.

The mean being substantially lower than the median suggests that lower-access observations pull the average downward.

Limited Water

Mean = 3.87%

Median = 0.47%

Mode = 0%

This indicates that limited water access is low for most observations, while a smaller number of observations have substantially higher values.

Unimproved Water

Mean = 4.42%

Median = 0.85%

Mode = 0%

Most observations have relatively low unimproved-water usage, but a smaller number of observations have much higher values.

Surface Water

Mean = 1.92%

Median = 0%

Mode = 0%

The zero median and mode indicate that surface-water dependence is absent or extremely low in many observations.

18. MEASURES OF SPREAD

The following were calculated:

- Q1
- Q3
- IQR
- Standard deviation

Results

Feature	Q1	Q3	IQR	Standard Deviation
Basic water	85.50%	100.00%	**14.50**	**15.11**
Limited water	0%	4.85%	**4.85**	**6.96**

Feature	Q1	Q3	IQR	Standard Deviation
Unimproved water	0.0339%	5.2793%	**5.25**	**7.17**
Surface water	0%	1.8783%	**1.88**	**3.92**

19. INTERQUARTILE RANGE

The IQR was calculated as:

IQR=Q3−Q1IQR=Q3-Q1

The IQR represents the spread of the **middle 50% of observations**.

Basic water

Q1 = 85.5%

Q3 = 100%

Therefore:

> IQR = **14.5 percentage points**

This indicates that the middle half of observations are concentrated in a relatively high basic-water-access range.

Limited water

IQR = **4.85 percentage points**

Most observations in the middle 50% have relatively low limited-water access.

Unimproved water

IQR = **5.25 percentage points**

The middle half remains relatively low, although there is greater variation outside the middle range.

Surface water

IQR = **1.88 percentage points**

This is the smallest IQR, indicating that surface-water dependence is tightly concentrated at low values for the middle 50% of observations.

20. STANDARD DEVIATION

Standard deviation was used to understand the overall variation of observations around the mean.

Results:

- Basic water — **15.11**
- Limited water — **6.96**
- Unimproved water — **7.17**
- Surface water — **3.92**

Key interpretation

“Basic water has the greatest absolute spread among the four service categories, based on both IQR (14.50 percentage points) and standard deviation (15.11 percentage points). Unimproved water has the largest standard deviation among the three lower-service categories, but its spread is occurring around a much lower mean.”

> **7.17**

This indicates comparatively greater variation in unimproved-water usage across observations.

Surface water has the smallest:

> **3.92**

indicating lower overall variation.

Corrected Interpretation of IQR and Standard Deviation

> **Basic water has the greatest absolute variation among the four service categories, with an IQR of 14.50 percentage points and a standard deviation of 15.11 percentage points. This indicates that both the middle 50% of observations and the overall distribution are more widely spread for basic-water access than for the other service categories.**

> **Among the three lower-service categories, unimproved water has the largest standard deviation (7.17) and IQR (5.25), followed by limited water and surface water. Although these values are smaller than those for basic water in absolute terms, the unimproved-water standard deviation is large relative to its low mean of 4.42%, indicating substantial variation around a generally low level.**
>
> **The fact that standard deviation is larger than IQR for each service category indicates that observations outside the middle 50% contribute additional spread. However, the difference between IQR and standard deviation alone should not be treated as proof of skewness or outliers; the distribution and other statistics should also be examined.**

Corrected comparison table

Variable	IQR	Standard Deviation	Correct interpretation
Basic Water	**14.50**	**15.11**	Greatest absolute variation
Limited Water	4.85	6.96	Lower absolute variation; some observations increase overall spread
Unimproved Water	5.25	7.17	Greatest spread among the three lower-service categories; large relative to its low mean
Surface Water	1.88	3.92	Lowest absolute variation

21. IQR VS STANDARD DEVIATION

The project provided an important opportunity to distinguish these two measures of spread.

IQR

Focuses on the middle 50%.

More resistant to extreme values.

Standard Deviation

Uses the entire set of observations.

More sensitive to unusually high or low values.

Professional lesson

When data are skewed or contain potential outliers, IQR can provide a more robust description of the central spread.

Standard deviation provides information about overall variability.

A strong analyst does not automatically choose one and ignore the other.

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22. BOX-AND-WHISKER / CANDLESTICK VISUALIZATION

A candlestick chart in Google Sheets was used to represent the box-and-whisker distribution.

The chart was built from the five-number summary.

Mapping

Candlestick element	Statistical measure
Low	Minimum
Open	Q1
Close	Q3
High	Maximum

The final analysis incorporated:

- 4 National features
- 4 Rural features
- 4 Urban features

for a total of:

> **12 water-access features**

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23. BOX-AND-WHISKER INTERPRETATION

The:

Whiskers

represent the broader observed range.

Box

represents the middle 50% of observations.

Lower edge of box

represents Q1.

Upper edge of box

represents Q3.

Box height

represents the IQR.

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24. KEY FINDING FROM AREA ANALYSIS

The distributions showed an important difference between urban and rural populations.

Urban

Basic water access is generally high and more concentrated toward the upper end.

Rural

Water-service levels display greater variability, with lower-level services appearing more visibly among some observations.

Overall implication

> **Water-access inequality appears more evident when the population is examined by area type, particularly between urban and rural populations.**

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25. INVESTIGATION 2 — ACCESS BY POPULATION SIZE

Objective

Investigate whether water-access levels appear to vary according to population characteristics.

Three perspectives were examined:

1. National population size
2. Urban population share
3. Rural population share

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26. NATIONAL POPULATION VS WATER ACCESS

A 100% stacked column chart was created.

X-axis

National population size.

Series

- Basic water
- Limited water
- Unimproved water
- Surface water

Purpose

To examine whether water-service composition changes as national population size changes.

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27. POPULATION-SIZE VISUALIZATION PROBLEM

The original population values were highly uneven.

Some countries had very small populations, while others had populations of hundreds of millions or over one billion.

As a result:

- The bars became very narrow.
- Labels became crowded.
- Outliers dominated the visual scale.
- The chart became difficult to interpret.

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28. OPTIONS CONSIDERED FOR OUTLIERS

Three approaches were considered.

Option A — Delete outliers

Advantage:

Cleaner chart.

Disadvantage:

Information is permanently lost.

Option B — Use an axis cutoff

Advantage:

Improves readability.

Disadvantage:

Part of the dataset disappears from the visualization.

Option C — Change the unit

Advantage:

Improves readability without deleting observations.

Disadvantage:

Reduces visual precision.

Project decision

> **Change the population unit for visualization purposes.**

---

29. AGGREGATION

Aggregation was then used to group observations into population categories.

When multiple countries share the same rounded population category, their water-access percentages can be summarized.

Aggregation method used

> **Average**

Why Average?

Water-access variables are percentages.

Adding percentages across multiple countries can produce meaningless values above 100%.

Average provides a representative percentage for the group.

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30. AGGREGATION — ANALYTICAL MEANING

Without aggregation:

> One bar can represent one country.

With rounding + aggregation:

> One bar can represent a group of countries within the same population category.

Therefore, aggregation reduces visual noise and helps reveal broad patterns.

---

31. ORDERING THE DATA

The x-axis initially appeared unordered because the underlying dataset was not arranged according to the independent variable.

The solution was to sort the underlying analytical data.

Examples:

National

Population size:

> Small → Large

Urban

`pop_u (rounded)`:

> 0 → 100

Rural

`pop_r (rounded)`:

> 0 → 100

Sorting the full dataset/analysis range was important so that the corresponding observations remained connected.

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32. URBAN POPULATION SHARE ANALYSIS

The project used:

> `pop_u`

as the urban population share.

A rounded feature:

> `pop_u (rounded)`

was created for grouping and visualization.

Aggregation was set to:

> **Average**

---

33. RURAL POPULATION SHARE ANALYSIS

Rural population share was derived from urban population share.

A rounded feature:

> `pop_r (rounded)`

was then created.

Aggregation was set to:

> **Average**

---

34. POPULATION-SIZE INVESTIGATION FINDINGS

National population

No simple, consistently increasing or decreasing relationship with water access was evident from the visualization.

Urban population share

Basic water remained the dominant service level across many groups.

Rural population share

Greater variation in water-service levels was more visible, with lower service levels appearing more prominently in some higher-rural-share groups.

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35. IMPORTANT ANALYTICAL LIMITATION

The project does **not** allow us to conclude:

> Population size causes water access to increase or decrease.

The charts show observed patterns/associations.

Other variables could also influence water access, including:

- Infrastructure
- Geography
- Economic resources

- Government investment
- Urbanization
- Population distribution
- Water-resource availability

Therefore:

> **Association should not be interpreted as causation.**

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36. INVESTIGATION 3 — ACCESS BY INCOME GROUP

Objective

Investigate the relationship between:

- Income group
- Population size
- Urbanization
- National water access

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37. INCOME GROUP CLASSIFICATION

The original income categories were:

- NAN
- Low income
- Lower middle income
- Upper middle income
- High income

A new numerical feature:

> `income_group_num`

was created.

---

38. NUMERICAL CODING

Code	Income group
0	NAN
1	Low income
2	Lower middle income
3	Upper middle income
4	High income

The codes were created primarily to provide a logical order for sorting and visualization.

They do **not** represent income magnitude.

For example:

> 4 does not mean High Income is four times Low Income.

It is an ordinal coding system.

---

39. NESTED IF FORMULA

A nested `IF()` function was used:

```
=IF(B2="NAN",0,
 IF(B2="Low income",1,
 IF(B2="Lower middle income",2,
 IF(B2="Upper middle income",3,
 IF(B2="High income",4,"")))))
```

A new analytical feature was created rather than overwriting the original income-group column.

Professional reason

The original category remains available while the analytical/ordering feature is separately documented.

---

40. PIVOT TABLE STRUCTURE

The pivot table was grouped by:

> `income_group_num`

Aggregations

Variable	Aggregation
`pop_n`	SUM
`pop_u`	AVERAGE
`wat_bas_n`	AVERAGE
`wat_lim_n`	AVERAGE
`wat_unimp_n`	AVERAGE
`wat_sur_n`	AVERAGE

---

41. WHY SUM FOR POPULATION?

Population is a count.

Therefore:

> SUM answers how many people are collectively represented within an income group.

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42. WHY AVERAGE FOR URBANIZATION AND WATER ACCESS?

Urban population share and water-access indicators are percentages.

Therefore:

> AVERAGE provides a representative country-level percentage for each income group.

Using SUM on these percentages could create values that exceed 100% and would not answer the intended question.

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43. INCOME-GROUP WATER ACCESS RESULTS

Income Group	Basic Water	Limited Water	Unimproved Water	Surface Water
Low income	**62.82%**	**16.55%**	**15.21%**	**5.42%**
Lower middle income	**82.21%**	**5.69%**	**7.90%**	**4.29%**
Upper middle income	**96.43%**	**1.56%**	**1.48%**	**0.57%**
High income	**99.56%**	**0.18%**	**0.24%**	**0.02%**

The NAN group was retained for data awareness but excluded from the main economic interpretation because it does not represent a genuine income category.

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44. INCOME GROUP — BASIC WATER FINDING

A very clear pattern was observed.

Basic water access increased across the income groups:

> **62.82% → 82.21% → 96.43% → 99.56%**

from:

> Low → Lower Middle → Upper Middle → High

Analytical insight

Higher-income groups in this dataset are associated with substantially higher average national basic-water access.

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45. LOWER WATER-SERVICE LEVELS

The opposite pattern appeared for:

Limited water

> 16.55% → 5.69% → 1.56% → 0.18%

Unimproved water

> 15.21% → 7.90% → 1.48% → 0.24%

Surface water

> 5.42% → 4.29% → 0.57% → 0.02%

As income category increases, these lower service levels generally decline.

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46. URBANIZATION BY INCOME GROUP

Average urban population share:

Income Group	Average Urban Share
Low income	**36.04%**
Lower middle income	**48.79%**
Upper middle income	**64.69%**
High income	**79.36%**

Finding

Urbanization generally increases as income category increases.

---

47. POPULATION BY INCOME GROUP

Total population represented in each group:

Income Group	Population (thousands)
Low income	**590,425.16**
Lower middle income	**3,399,310.23**
Upper middle income	**2,547,619.31**
High income	**1,212,076.07**

Finding

Population does not increase consistently as income group increases.

The lower-middle-income group contains the largest total population represented in the dataset.

Therefore:

> **Income group and total population do not show a simple linear relationship.**

---

48. OVERALL INCOME-GROUP INTERPRETATION

Three broad patterns emerged:

Income ↔ Water Access

Higher income groups generally show higher basic-water access.

Income ↔ Urbanization

Higher income groups generally show greater urban population shares.

Income ↔ Population

Population size does not show the same straightforward progression.

This distinction is important because it prevents us from assuming that income, population and urbanization behave in the same way.

Across the investigations, several consistent themes emerged.

Finding 1 — Basic Water Access is generally dominant

Basic water is the largest service level in many observations.

Finding 2 — Rural populations show greater variation

Urban areas generally show stronger and more consistent basic-water access than rural areas.

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49. OVERALL PROJECT FINDINGS

Finding 3 — National population size alone does not explain water access

The population-size visualization did not show a simple monotonic pattern.

Finding 4 — Population structure provides additional insight

Urban/rural population share appears more informative than population size alone in understanding differences in access.

Finding 5 — Income group shows a strong relationship with water access

Basic water access increases markedly across the income categories.

Finding 6 — Lower-level services decline as income increases

Limited, unimproved and surface water levels generally decrease as income group rises.

Finding 7 — Urbanization and income move together in this dataset

Average urban population share generally increases across the income groups.

50. CAUSATION WARNING

The analysis identifies relationships and patterns.

It does **not** establish that:

> Income causes better water access.

or:

> Rural population share causes poorer water access.

Further analysis would be necessary to investigate possible causal explanations.

Potential factors for future investigation include:

- Infrastructure

- Government investment
- Economic conditions
- Geography
- Population density
- Water-resource availability
- Urban planning
- Rural remoteness

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51. PROFESSIONAL DATA-ANALYSIS LESSONS

This project developed several important analyst skills.

Data Quality

I learned to question:

- Missing values
- Impossible percentages
- Incorrect imports
- Data types
- Structural inconsistencies

Statistical Analysis

I applied:

- Mean
- Median
- Mode
- Minimum
- Maximum
- Quartiles

- IQR
- Standard deviation

Spreadsheet Analysis

I used:

- `COUNTA()`
- `COUNT()`
- `SUM()`
- `AVERAGE()`
- `MAX()`
- `MIN()`
- `ROUND()`
- `ROUNDUP()`
- `ABS()`
- `MEDIAN()`
- `MODE()`
- `QUARTILE()`
- `STDEV()`
- Nested `IF()`

Data Visualization

I worked with:

- 100% stacked column charts
- Box-and-whisker/candlestick charts
- Population-group visualizations
- Income-group comparisons

Analytical Thinking

I learned to:

- Question the data.
- Question the visualization.
- Question the analytical variable.
- Distinguish population size from population share.
- Distinguish counts from percentages.
- Choose SUM versus AVERAGE according to the meaning of the variable.
- Recognize patterns without automatically claiming causation.
- Consider the stakeholder before choosing a visualization.

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52. PROFESSIONAL CRITIQUE OF THE PROJECT METHODS

The project methodology is useful for learning exploratory analysis, but several methods require careful communication in professional settings.

Population rounding

Rounding small populations to the nearest whole million can create large distortions.

Therefore, rounded population values should be used for grouping/visualization rather than precise reporting.

Averaging percentages

The project uses simple averages across countries.

This means a small country and a large country can contribute equally to the group average.

A future professional analysis could explore **population-weighted averages** if the business question is about the experience of people rather than the typical country.

Income coding

The 1–4 income codes create order, but they should not be treated as continuous numerical measurements.

Aggregation

Aggregation improves readability but hides individual-country variation.

Therefore, aggregated charts should be complemented with detailed country-level analysis when important decisions are being made.

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53. STAKEHOLDER / NGO INTERPRETATION

For an NGO or policymaker, the most useful message is not simply:

> "The average is 82.21%."

The analyst should translate the statistic into meaning:

> "The lower-middle-income group has substantially higher average basic-water access than the low-income group, but a measurable gap remains."

Likewise:

> "Higher rural population shares are associated with greater variability in water-service levels, suggesting that rural communities may require more targeted investigation."

The role of the analyst is to translate:

Data → Information → Insight → Decision

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54. RECOMMENDED DECISION-MAKING FRAMEWORK

When considering an intervention, percentage alone should not determine the decision.

A stronger framework considers:

Severity

How bad is the access level?

Scale

How many people are affected?

Vulnerability

Who is affected?

Risk

Are people relying on unimproved or surface water?

Feasibility

What intervention is practical?

Cost-effectiveness

How much improvement can be achieved with available resources?

This is an important professional expansion beyond the portal.

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55. PERSONAL ANALYST LEARNING

This project changed my approach to data analysis.

Initially, it was easy to think that the main task was:

> Calculate the numbers and create the charts.

During the project, I learned that analysis requires more than calculation.

I learned to ask:

> What does this variable actually represent?

> Is this value reasonable?

> Does the chart communicate the information clearly?

> Am I comparing percentages or actual counts?

> What does aggregation do to the meaning?

> Is this relationship causal or simply an observed association?

> Would a stakeholder understand this visualization?

This shift from **tool usage to analytical reasoning** is one of the major learning outcomes of the project.

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56. PROJECT LIMITATIONS

Several limitations should be acknowledged.

1. Descriptive analysis

The project primarily establishes descriptive relationships.

2. Aggregation

Country-level variation can be hidden when observations are grouped.

3. Simple averages

Averages may not represent the experience of the average person if population sizes differ greatly.

4. Rounded visualization values

Rounding can reduce precision.

5. Potential data-quality limitations

Some values required cleaning or correction.

6. No causal modelling

The project does not prove why differences in water access exist.

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57. PORTFOLIO VALUE

This project demonstrates ability in:

- Data preparation
- Data validation
- Spreadsheet analysis
- Descriptive statistics
- Exploratory analysis

- Pivot tables
- Data visualization
- Business questioning
- Interpretation
- Data-quality assessment
- Analytical storytelling

The strongest portfolio angle is not:

> "I created charts in Google Sheets."

It is:

> **"I investigated a real-world water-access problem, validated the dataset, analyzed population and socioeconomic patterns, identified differences in urban and rural access, and translated the results into decision-oriented insights."**

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58. RECOMMENDED PROJECT SCREENSHOTS

Keep the following screenshots as your project evidence:

Screenshot 1

Data validation using `value_cnt`.

Screenshot 2

Corrected rows after splitting semicolon-separated values.

Screenshot 3

Global 2020 Report — population comparison.

Screenshot 4

Urban vs Rural population visualization.

Screenshot 5

National water-access descriptive statistics.

Screenshot 6

National box-and-whisker/candlestick chart.

Screenshot 7

Combined National/Urban/Rural water-access distribution chart.

Screenshot 8

National population vs water-access chart.

Screenshot 9

Urban population-share vs water-access chart.

Screenshot 10

Rural population-share vs water-access chart.

Screenshot 11

Income-group pivot table.

Screenshot 12

Final income-group water-access visualization.

Screenshot 13

Urbanization by income group.

Screenshot 14

Population by income group.

For the **final portfolio presentation**, we should not necessarily show all 14. We can select the strongest 5–7.

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59. FINAL PROJECT CONCLUSION

> **This project investigated access to drinking water across countries using population characteristics, urban and rural population structure, and income group. The analysis began with data validation and population**

> **coverage assessment before moving into descriptive statistics and visualization.**
>
> **The dataset represented approximately 99.56% of the estimated 2020 world population, providing strong population coverage for the analysis. The water-access investigation showed that basic water services were generally the dominant service level, although substantial variation existed between observations and particularly across urban and rural areas.**
>
> **Population-size analysis did not reveal a simple consistent relationship between national population size and water access. However, area type and population structure provided additional insight, with rural populations showing greater variability in water-service levels.**
>
> **The income-group analysis revealed the clearest pattern: average basic-water access increased substantially from low-income to high-income groups, while limited, unimproved and surface-water access generally decreased. Urbanization also increased across the income groups, while total population did not follow a simple income-based progression.**
>
> **Overall, the project demonstrates that water access is influenced by a combination of population structure, economic conditions and other contextual factors. The analysis provides evidence of important relationships but does not establish causation. Further investigation would be required to identify the underlying drivers and determine the most effective interventions.**

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60. CORE PROJECT TAKEAWAY

> **Data analysis is not simply about finding numbers. It is about validating the data, choosing appropriate methods, understanding what the numbers represent, recognizing patterns, questioning assumptions, communicating findings clearly, and knowing what the evidence does—and does not—allow us to conclude.**

DOCUMENT 1 STATUS

Official Overall Project Documentation: ✅ Draft consolidated

This will serve as the **professional/master version** of the project.

The second document will be completely different in style:

DOCUMENT 2 — PROJECT WALKTHROUGH / DATA STORY

That one will follow the journey as a story:

“Can we trust this dataset?”

→ **“How much of the world does it represent?”**

→ **“Who has access to water?”**

→ **“Does location matter?”**

→ **“Does population matter?”**

→ **“Does income matter?”**

→ **“What did the data teach us?”**

That version can then be adapted into your **YouTube presentation, Facebook/Instagram story, LinkedIn project presentation, and eventually your broader data-solutions brand narrative**.

This official document should be the source of truth from which those public-facing versions are developed.